

happen automatically and in real time.

A single RAMCloud system can be used to support multiple applications of varying sizes. It should provide a security mechanism to support mutual hostile applications. Also, one application with a very high workload must not degrade the performance of other applications.

Finally, RAMClouds must manage themselves automatically. There are thousands of servers, each using hundreds of peers, which makes the overall design too complicated to be handled by humans.

Why use RAMCloud?

We believe there are two main motivations for using RAMCloud.

Application scalability

Most Web applications use relational databases to store their data. As the application grows, it becomes difficult to store the entire data in a single relational database. Applications then use other techniques to manage their data. A popular technique is 'Ad-hoc', where data is partitioned among multiple databases. As the application grows larger, maintaining consistency among multiple databases becomes increasingly difficult and requires more complex techniques to overcome these issues. Another storage technique is 'Bigtable', which is built on top of a Google file system. Because of the distributed nature of a Bigtable database, performing certain database operations like a join between two tables would be terribly inefficient. On the other hand, RAMCloud will automatically scale to support the increasing number of storage servers used by an application.

The technology

The disk is used as a primary storage system for Web applications. Accessing large blocks at one time from a disk may be beneficial when compared to accessing small blocks. However, most forms of online data, such as images and songs, do not comprise large blocks. So the latency for accessing the smaller more frequent blocks of memory is high. Large Web applications need to make multiple internal requests to generate a single HTML page. So we need to consider the cumulative latency of all the requests while considering the overall response time to users. One of the major advantages of RAMCloud over the disk-oriented approach is that it can dramatically reduce the access latency of a request and, thereby, reduce the overall response time. RAMCloud also supports a new class of data-intensive applications, which process data in large volumes—typically, in terabytes.

The pros and cons

Here are some of the pros and cons of using RAMCloud.

The pros

1. Since all the information is stored in DRAM, RAMCloud provides high throughput.
2. RAMCloud automatically scales to support a large number of storage servers and eliminates the scalability issues in applications.
3. It provides high level of data durability and availability.
4. The cost of storing data on DRAM today is the same as storing data on a disk ten years ago.
5. RAMCloud supports a log-structure similar to a log structured file system for all its data on DRAM as well as on disk. This provides fast crash recovery.
6. RAMClouds are 100-1000x more efficient than disk-based systems in terms of cost per operation or energy per operation.

The cons

1. It involves a higher cost/bit and high energy/bit, so RAMCloud storage will be 50-100x more expensive than a pure disk-based system.
2. Maintaining consistency for applications that require replication across data centres is very difficult.

With the growth of large scale Web applications, there has been a need for alternative disk storage technologies. Both Google and Yahoo store their search indices entirely in DRAM. The Bigtable storage system allows entire column families to be loaded into memory, where they can be read without any disk accesses. We believe that RAMCloud is a long-term solution for the storage needs of Web applications. RAMCloud provides durability and very low latency. Hence, it enables richer query models and is attractive for technologies like cloud computing. It is able to aggregate the resources of a large number of commodity servers. However, a lot of research needs to be done and numerous challenges must be overcome in order to use this technology. 

References

- [1] <https://ramcloud.stanford.edu>
- [2] <http://www.stanford.edu/~ouster/cgi-bin/papers/ramcloud.pdf>

Acknowledgements

I would like to thank Dr John Ousterhout, professor of Computer Science at Stanford University. He is the lead at the RAMCloud project at Stanford University. I would also like to thank my mentor and all the people who helped me to review this article.

By: Sakshi Bansal

The author is in her fourth year of the Computer Science and Engineering bachelors' degree at Amrita Vishwa Vidyapeetham, Amritapuri. She is a FOSS enthusiast and an active member of the Amrita FOSS club, having made contributions to various open source projects such as Mozilla Thunderbird, Mediawiki, etc. She blogs at <http://sakshiil.wordpress.com/>.